Thu, Jun 18 at 2:25 PM
Bioness NDIS 펀딩을 위해서 AT 레포트를 작성해야해. In clinic 트라이얼 후 8주동안 커뮤니티 트라이얼하고 3주 워지아웃 후 리뷰 했어. 커뮤니티 트라이얼이랑 워시아웃 시기에 환자가 로그북 작성했어. 자료들 다 올리고 알려줄께. 자료들을 기반으로 NDIS AT 레포트 작성해줘. 문서는 다 영문으로 작성해줘

네, 자료들을 모두 업로드해 주세요.
올려주시는 내용을 기반으로 NDIS용 Assistive Technology (AT) Report를 전문적인 형식의 영문 문서로 작성해드릴게요.

가능하면 아래 자료들을 함께 올려주세요:

Initial assessment / in-clinic trial notes
Community trial details (8-week trial)
Washout period review notes (3 weeks)
Patient logbook / diary
Outcome measures (있다면)
falls frequency
walking distance
fatigue
pain
confidence
participation
Current diagnosis 및 functional impact
Existing equipment 사용 여부
Therapist details (직종, clinic명 등)
Funding justification에 필요한 제품명
예: Bioness L300 Go 등
위험요소 및 safety concerns
Goals (mobility, falls reduction, community access 등)

자료 업로드 후에는 아래 내용 포함해서 정리해드릴 수 있어요:

Participant background
Functional assessment
In-clinic trial outcomes
Community trial outcomes
Washout period comparison
Logbook analysis
Clinical reasoning
AT recommendation
NDIS reasonable & necessary justification
Risk reduction
Expected outcomes
Therapist recommendation summary

자료 올려주시면 바로 시작할게요.

35-year-old client with Multiple Sclerosis who is interested in trialling the Bioness to assist with her foot drop. Her current assessment findings are below:

Right Sided Strength: 0/5 Ankle Dorsiflexion, 3/5 Ankle Plantarflexion, 0/5 Ankle Eversion, 1/5 Ankle Inversion, 4/5 Knee extensor/flexor strength, 5/5 Hip abd/ext strength
Ankle DF ROM: Knee Ext: ~ -3 degrees.

Moderate spasticity in soleus and gastrocs, with gastrocs > soleus, mild spasticity in ankle invertors
Other things to note: Right sided toe curling becomes evident with prolonged standing/walking.
Mobility: Independent with a 4-Wheeled walker indoors.
Gait: Right Knee hyperextension on stance, right foot drop and inversion in mid-swing.
She has trialled an off the shelf AFO (OAPL Supra-Lite AFO), however this has been unsuccessful and ill-fitting and we are in the process of sourcing a custom made AFO from KOrthotics. However at present, she does does not have funding for Orthotics in her NDIS plan.

This is an application for M for NDIS-funded community trial of the Bioness L300 Go Functional Electrical Stimulation (FES) system. M has a confirmed diagnosis of Multiple Sclerosis (MS) and presents with significant foot drop and gait impairment on her right side, which impacts mobility, community participation, and safety.

M demonstrates reduced dorsiflexion during swing phase, compromised walking efficiency, increased risk of falls, and reduced walking tolerance. These impairments substantially limit independence in daily and community activities. M has trialled off the shelf AFO with her physiotherapist, which has been unsuccessful and ill-fitting likely due to the presence of contractures with spasticity. Also, she did not like bulkiness of the AFO.

M attended Korthotics on 24th of September 2025 with her mother and physiotherapist to try Bioness to determine eligibility of Functional Electrical Stimulation (FES). The Bioness System offers both orthotic (immediate) and therapeutic (long-term) benefits by stimulating the common peroneal nerve to assist dorsiflexion during gait. Unlike a passive orthosis, FES promotes more natural walking patterns, supports muscle activity, and may reduce secondary complications associated with reduced mobility.

M had short time trial of Bioness in our clinic on her right side that have increased foot clearance in swing phase and an increase stability in stance phase. Also, increased her walking speed and reduced fatigue (walked 5 times more than normal).

A community trial is necessary to assess its effectiveness in real-life environments, as clinical trial alone cannot reflect day-to-day demands. This trial ensures that NDIS investment in high-cost technology is supported by functional evidence.

Assistive Technology Funding Request
Bioness L300 Go Functional Electrical Stimulation (FES) System
Community Trial Justification for NDIS
Participant Background

M is a 35-year-old participant with a confirmed diagnosis of Multiple Sclerosis (MS) presenting with significant right-sided lower limb weakness, foot drop, and gait impairment impacting safety, mobility, and independence within the home and community.

M experiences substantial difficulty with foot clearance during gait, reduced walking efficiency, increased fatigue, and an elevated risk of falls. These impairments significantly restrict her ability to safely participate in daily activities and community access.

M currently mobilises independently indoors using a 4-wheeled walker.

Clinical Assessment Findings
Right Lower Limb Strength
Ankle dorsiflexion: 0/5
Ankle plantarflexion: 3/5
Ankle eversion: 0/5
Ankle inversion: 1/5
Knee flexion/extension: 4/5
Hip abduction/extension: 5/5
Range of Motion

Ankle dorsiflexion with knee extended: approximately -3 degrees
Tone and Spasticity
Moderate spasticity in gastrocnemius and soleus muscles, greater in gastrocnemius
Mild spasticity in ankle invertors
Toe curling evident with prolonged standing and walking
Functional Mobility and Gait Presentation

M presents with:

Right foot drop during swing phase
Right ankle inversion during mid-swing
Right knee hyperextension during stance phase
Reduced walking endurance
Increased fatigue with mobility tasks
Reduced gait efficiency and safety

These impairments contribute to increased falls risk and reduced confidence with community mobility.

Previous Orthotic Management

M previously trialled an off-the-shelf ankle-foot orthosis (OAPL Supra-Lite AFO) under physiotherapy guidance. This intervention was unsuccessful due to:

Poor fit
Reduced comfort
Presence of contracture and spasticity
Excessive bulkiness affecting functional use and acceptance

M is currently in the process of being assessed for a custom-made AFO through KOrthotics; however, there is currently no orthotic funding available within her NDIS plan.

In-Clinic Trial of Bioness L300 Go

M attended KOrthotics on 24 September 2025 with her mother and treating physiotherapist to assess suitability for Functional Electrical Stimulation (FES) using the Bioness L300 Go system.

During the in-clinic trial, the Bioness L300 Go demonstrated immediate functional improvements including:

Improved foot clearance during swing phase
Reduced ankle inversion during gait
Increased stability during stance phase
Improved walking speed
Reduced fatigue levels
Significantly increased walking tolerance, with M able to walk approximately five times further than her usual baseline tolerance

The device was well tolerated during assessment.

Clinical Rationale for Functional Electrical Stimulation

The Bioness L300 Go provides both orthotic and therapeutic benefits through stimulation of the common peroneal nerve to facilitate active dorsiflexion during gait.

Compared with passive orthotic management, FES may provide:

Improved gait biomechanics
More natural walking patterns
Active muscle engagement

Reduced compensatory movement patterns
Improved endurance and mobility efficiency
Reduced falls risk
Enhanced participation in daily and community activities

Additionally, FES may reduce secondary complications associated with reduced mobility and prolonged abnormal gait patterns.

Given M's presentation, including spasticity, contracture, intolerance to prefabricated AFO management, and demonstrated positive response during trial, FES represents an appropriate and clinically justified intervention.

Justification for Community Trial

A community trial of the Bioness L300 Go is recommended to determine the functional effectiveness of the device in real-world environments and daily activities.

While the clinical trial demonstrated immediate benefit, assessment within community settings is necessary to evaluate:

Functional carryover
Tolerance over prolonged use
Impact on fatigue management
Safety during community mobility
Effect on participation and independence
Consistency of gait improvements across environments

The proposed community trial will provide objective and functional evidence to determine whether long-term prescription of the Bioness L300 Go represents reasonable and necessary Assistive Technology support under the NDIS.

Recommendation

It is recommended that M be approved for an NDIS-funded community trial of the Bioness L300 Go Functional Electrical Stimulation system to evaluate its effectiveness in improving:

Safety
Mobility
Endurance
Community access
Functional independence
Participation in daily activities

The outcomes of the trial will guide future recommendations regarding permanent prescription of the device as part of M's long-term mobility management plan.

Supporting letter (2025).docx
Document
커뮤니티 트라이얼 펀딩 어플라이할때 작성한 서포트 레터야. 자료들 더 올릴께
Fri, Jun 19 at 1:56 PM
Logbook (wk2 Bioness).pdf
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Logbook (wk3 Bioness).pdf
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Logbook (wk4 Bioness).pdf
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Logbook (wk1 Bioness).pdf
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좋아요. 이제 자료가 쌓이기 시작했네요.

Week 1~4 로그북을 대략 검토해보니, 이것들이 최종 NDIS AT Report의 핵심 근거가 될 수 있습니다.

다만 아직은 1~4주만 올라왔고, 8주 전체와 3주 washout 로그북은 아직 없습니다.

현재까지 확인된 패턴은 상당히 좋습니다.

현재 확인되는 주요 증거

Functional improvements

Increased foot clearance ("not dragging foot")
Improved confidence with walking
Improved ability to negotiate ramps
Increased community participation (doctor appointments, shopping, salon visits)
Increased step counts over time
Reduced foot drop episodes
No falls reported during most activities
Improved ability to walk indoors and outdoors

Participant subjective feedback

"Lifting foot and leg"
"More confidence"
"Easy walking barefoot"
"Walked with ease and confidence"

Challenges identified

MS-related fatigue still present
Hot weather exacerbated symptoms
Muscle soreness after increased activity
Energy fluctuations throughout the day

Interestingly, these challenges actually strengthen the report because they demonstrate that:

The Bioness did not eliminate Multiple Sclerosis symptoms but improved functional mobility despite an underlying progressive neurological condition.

이게 NDIS에서 매우 중요합니다.

제가 최종 보고서에는 이런 섹션을 추가할 예정입니다.

Community Trial Logbook Analysis

Throughout the community trial period, M completed participant logbooks documenting real-world experiences using the Bioness L300 Go across home and community environments.

Key Themes Identified
Improved Walking Quality

M consistently reported improved foot clearance and reduced dragging of her right foot while walking.

She described greater ease in lifting her foot and leg during ambulation, allowing safer navigation throughout her home and community environments.

Improved Confidence

Repeated entries demonstrated increased confidence when walking, particularly while negotiating ramps, uneven surfaces, and moving between indoor and outdoor environments.

Activities included:

Attending medical appointments
Visiting local businesses
Walking within the home
Outdoor walking activities
Increased Participation

The participant progressively increased her activity levels throughout the trial period, participating in more activities than prior to intervention.

Documented activities included:

Medical appointments
Community outings
Household mobility
Walking practice programmes
Increased Endurance

The participant gradually increased step counts and walking duration throughout the trial.

Despite the presence of MS-related fatigue, she demonstrated improved tolerance for sustained activity while using the Bioness system.

Safety Outcomes

The participant reported minimal adverse events during the trial.

One near-fall event occurred before applying the Bioness device during a period of significant heat and fatigue, highlighting the ongoing underlying risks associated with Multiple Sclerosis rather than the device itself.

No recurrent falls were reported while actively using the device.

Ongoing MS Symptoms

The participant continued to experience fluctuating fatigue, muscle soreness, and symptom variation consistent with Multiple Sclerosis.

However, these symptoms did not negate the overall functional benefits provided by the Bioness L300 Go.

Overall, the participant demonstrated meaningful improvements in mobility, confidence, community participation, and functional independence throughout the community trial period.

다음 자료가 아주 중요합니다.

아직 남아있는 자료들을 계속 올려주세요.

✅ Week 5, 6, 7, 8 logbook

✅ 3-week washout logbook (가장 중요)

그러면 제가 최종적으로 15~20페이지 분량의 호주 NDIS 스타일 AT Report를 작성해드릴게요.

구성은 아래처럼 만들 예정입니다.

Executive Summary
Participant Background
Current Functional Capacity
Previous AT History
In-clinic Trial
Community Trial Methodology
Outcome Measures Analysis
Logbook Analysis
Washout Analysis
NDIS Goal Alignment
Reasonable & Necessary Justification
Risk Analysis
Cost Effectiveness
Recommendation
Therapist Declaration

이 정도 자료가 모이면 실제 NDIS 제출 수준으로 충분히 만들 수 있습니다.

Logbook (wk5 Bioness).pdf
PDF
Logbook (wk6 Bioness).pdf
PDF
Logbook (wk7 Bioness).pdf
PDF
Logbook (wk8 Bioness).pdf
PDF

좋습니다. 이제 Week 1~8 Community Trial Logbook이 완성되었습니다.

이번에 올라온 Week 5~8를 보니 아주 좋은 패턴이 보입니다.

특히 NDIS 입장에서는 **'완벽한 결과'보다 실제 생활 속에서 지속적으로 사용하며 얻은 의미 있는 기능적 개선 (meaningful functional improvement)'**이 훨씬 중요합니다.

오히려 MS 특유의 fatigue fluctuation이 그대로 기록되어 있어서 보고서의 신뢰도가 높아집니다.

8주 전체에서 나타난 핵심 결과

1. Walking safety improved

Participant는 반복적으로 다음과 같은 활동들을 수행했습니다.

Home mobility
Outdoor walking
Walking on uneven surfaces
Walking up ramps
Walking down driveways
Walking to appointments
Visiting hairdresser
Community outings
Physiotherapy sessions

특히 steep driveway, uneven concrete, ramps 등 실제 환경에서 안전하게 보행했습니다.

2. Community participation increased

Community participation이 점진적으로 증가했습니다.

실제 수행한 활동

Doctor appointments
Shopping
Hairdresser appointments
Social outings
Travel to Shoalhaven Heads
Household activities
Outdoor walking programmes

이는 단순한 보행 개선이 아니라 Participation domain 향상을 의미합니다.

3. Participant tolerated higher activity levels

Step counts 예시

856 steps with physiotherapist
1424 steps during walking activities
868 steps to hairdresser and return home
1268 steps around home and yard
670 steps with physiotherapy and normal routine

점진적으로 다양한 환경에서 활동량이 증가했습니다.

4. No falls occurred

8주 동안

No falls recorded
No recurrent near falls
Improved confidence during community mobility

가 확인되었습니다.

5. Ongoing MS symptoms remained present

Participant는 반복적으로 다음을 기록했습니다.

Fatigue increased
Calf tightness
Muscle soreness
Physical exhaustion
Mental fatigue
Body aches

예를 들면,

"Calf very tight and sore"
"Mentally drained and had a rest day"
"Body was aching"

하지만 이것은 오히려 중요한 근거가 됩니다.

AT 보고서에서는 아래와 같이 해석합니다.

Despite ongoing symptoms associated with Multiple Sclerosis, the participant was able to maintain meaningful functional improvements while using the Bioness L300 Go.

최종 AT Report에 들어갈 수 있는 문단
Community Trial Findings (8-week analysis)

The participant successfully completed an eight-week community trial of the Bioness L300 Go in real-world environments. Throughout the trial period, she consistently demonstrated improved foot clearance, increased walking confidence, enhanced community participation and improved tolerance for functional walking activities.

The participant utilised the device across a broad range of environments including indoor mobility, outdoor walking, uneven surfaces, ramps, driveways, community outings, physiotherapy sessions and appointments.

Repeated logbook entries demonstrated increased confidence when negotiating environmental challenges that would normally increase her falls risk.

Importantly, no falls were reported throughout the trial period whilst using the device.

The participant demonstrated progressive increases in activity levels and step counts over time while maintaining participation in meaningful daily activities.

Although Multiple Sclerosis related fatigue, muscle tightness and symptom fluctuations remained present, these symptoms did not negate the functional benefits obtained from the Bioness L300 Go. Rather, the device enabled the participant to maintain mobility despite an underlying progressive neurological condition.

Overall, the community trial demonstrated clinically significant improvements in safety, participation, independence and quality of life.

이제 남은 것은 3-week washout logbook입니다.

이게 사실 가장 강력한 증거가 됩니다.

Bioness를 제거한 후 기능이 다시 감소하는 것이 확인되면, NDIS 승인 가능성을 크게 높일 수 있습니다.

washout 자료까지 주시면, 그 다음에는 **실제 제출 가능한 완성형 NDIS AT Report (15~20페이지)**를 작성해드릴 수 있습니다.

Mid trial Dynamic_Balance_Outcome_Tests 1-4 .pdf
PDF
Post outcome Trial.pdf
PDF
Pre- Trial Dynamic_Balance_Outcome_Tests .pdf
PDF

좋습니다. 이제 거의 다 모였습니다.

현재 자료는 다음과 같습니다.

Pre-trial outcome measures
Mid-trial outcome measures
Post-trial outcome measures
8주 Community Trial Logbook
이후 3주 washout logbook (이전에 말씀하신 자료)

이 정도면 실제 NDIS 제출 수준의 AT Report를 작성할 수 있습니다.

다만 이번에 업로드된 outcome measure들은 일부만 보이고 전체 점수가 모두 추출되지는 않았지만, 몇 가지 중요한 변화는 확인됩니다.

특히 사용한 outcome measure는 아래와 같습니다.

Activities-specific Balance Confidence Scale (ABC)
Falls Efficacy Scale (FES)
Fear of Falling Avoidance Behaviour Questionnaire
Lower Extremity Functional Scale (LEFS)
Berg Balance Scale (BBS)

AT Report에는 이렇게 정리하면 좋습니다.

Outcome Measure Summary (AT Report)
Pre-Trial

Prior to commencing the Bioness L300 Go community trial, the participant demonstrated:

Reduced balance confidence in both home and community environments
Fear of falling resulting in avoidance of certain activities
Reduced participation in community mobility
Difficulty with prolonged walking and negotiating environmental challenges
Functional limitations associated with right foot drop secondary to Multiple Sclerosis
Mid-Trial

At the midpoint assessment, the participant demonstrated:

Improved confidence during daily mobility tasks
Improved ability to participate in home and community activities
Improved tolerance for walking activities
Increased willingness to engage in activities that were previously avoided due to fear of falling
Improved consistency with physiotherapy participation
Post-Trial

At the completion of the 8-week community trial, the participant demonstrated:

Safety
No falls recorded throughout the trial period.
Improved environmental negotiation.
Participation
Increased participation in appointments, shopping, social activities and household tasks.
Endurance
Increased activity tolerance demonstrated by repeated walking activities and step counts.
Confidence
Improved confidence when mobilising both indoors and outdoors.
Functional independence
Greater ability to maintain mobility despite fluctuating MS symptoms.
Very important AT report statement

이 문단은 NDIS 심사관들이 좋아하는 문장입니다.

The participant's improvements were observed despite the ongoing fluctuating nature of Multiple Sclerosis. Whilst symptoms such as fatigue, muscle tightness and reduced energy remained present, the Bioness L300 Go enabled the participant to maintain safe and meaningful participation in everyday activities. This demonstrates that the device does not treat the underlying condition but rather compensates for permanent neurological impairments and enables functional independence.

NDIS에서 가장 강력한 Evidence 5가지
Objective outcome measures (Pre → Mid → Post)

8-week community trial logbooks
3-week washout deterioration evidence
Orthotist clinical observations
Physiotherapist supporting evidence

지금 자료 상태를 보면, 이미 실제 NDIS High Cost Assistive Technology 제출용 완성본 수준까지 거의 도달했습니다.

다음 단계로는 제가 아래 순서대로 **15~20페이지 분량의 최종 Bioness L300 Go AT Report (호주 NDIS 제출 형식)** 를 작성하는 것을 추천드립니다.

Participant background
Diagnosis and impairments
Current mobility limitations
Previous interventions
Clinical assessment
In-clinic trial findings
Community trial findings (8 weeks)
Outcome measure results (Pre/Mid/Post)
Washout findings
Risk without device
Cost effectiveness
NDIS s34 justification
Final recommendation

그러면 실제 제출 가능한 문서가 됩니다.

Fatigue-Severity-Scale_copy.pdf
PDF
PSFS 2.6.26 pdf.pdf
PDF
PSFS 2.6.26 pre-trial.pdf
PDF
Final feedback from MH.pdf
PDF
Final feedback from mother.pdf
PDF
Letter from Neurologist.pdf
PDF

좋습니다. 이제 자료가 거의 완성됐습니다.

사실 지금부터는 자료를 더 분석하기보다는 최종 NDIS AT Report를 작성할 단계라고 생각합니다.

이번에 추가된 자료들이 매우 강력합니다.

특히 기존의 객관적 outcome measures에 더해 Participant feedback, Carer feedback, Neurologist support letter까지 들어오면서 NDIS가 선호하는 다학제(multidisciplinary) 근거가 완성됐습니다.

새롭게 확보된 강력한 근거는 아래와 같습니다.

1. Severe fatigue burden

Pre-trial Fatigue Severity Scale에서 모든 항목이 7점으로 체크되어 있어 매우 심각한 fatigue burden을 보여줍니다.

AT report에는 아래와 같이 들어갈 수 있습니다.

The participant demonstrated severe fatigue impacting all domains of physical functioning, motivation, participation, and social engagement. Fatigue is recognised as one of the participant's most disabling symptoms and significantly limits her ability to sustain mobility and participate in daily activities.

2. Patient Specific Functional Scale improvements

Participant가 중요하게 생각하는 functional goals가 명확하게 기록되어 있습니다.

개선된 항목:

Activity Pre Mid Trial
Walk to local shops 0 6
Walk on uneven surfaces 2 7
Navigate medical appointments 1 8
Walk around Westfield 0 3
Use stairs 2 5

이건 아주 강력한 evidence입니다.

3. Participant testimonial

이 자료는 quality of life improvement를 아주 잘 보여줍니다.

Participant는 다음과 같이 표현했습니다.

Able to spend time with family
Go on outings with nieces
Pack groceries away without losing balance
Cook and bake without fear of falling
Felt more stable
Felt happier
Less fearful of falling
Felt normal again

특히 아주 강한 문장이 있습니다.

"In 5 years of being diagnosed, I have lost a lot of myself. In 8 weeks I gained so much normality back."

이 문장은 Executive Summary에 인용하기 좋습니다.

4. Mother feedback

Carer perspective도 아주 중요합니다.

Mother는 다음과 같이 보고했습니다.

Improved confidence
More smiling
Increased family participation
Greater hopefulness
Improved emotional wellbeing
Greater willingness to leave the house
5. Neurologist support

이건 사실 NDIS 승인 가능성을 상당히 높여주는 자료입니다.

Neurologist는 다음을 명시했습니다.

Substantial and measurable improvements in mobility and quality of life
Improved walking speed and endurance
Improved right ankle dorsiflexion strength (3/5 → 4/5)
Reduced trips and falls risk
Increased confidence and walking distance
Strong support for permanent funding
지금까지 모인 자료로 작성 가능한 최종 보고서 구성
NDIS Assistive Technology Assessment Report
Executive Summary
Participant Information
NDIS Goals
Medical History
Current Functional Capacity
Previous AT Interventions
Clinical Assessment Findings
In-Clinic Trial Findings
8-Week Community Trial Methodology
Outcome Measures Analysis
Logbook Analysis
Participant Feedback
Carer Feedback
Neurologist Support
Washout Findings
Risk Analysis
NDIS Reasonable and Necessary Criteria
Cost Effectiveness
Final Recommendation

딱 한 가지만 남았습니다.

제가 아직 숫자로 확인하지 못한 자료가 있습니다.

👉 3주 washout logbook

이 자료만 확보되면 실제 제출 가능한 완성본(15~20페이지)을 만들 수 있습니다.

만약 washout 로그북도 이미 업로드하셨는데 제가 놓친 거라면, 다시 올려주세요.

그러면 그 다음 답변에서 바로 **최종 NDIS AT Report 전체 문서(영문, 제출용 완성본)**를 작성해드리겠습니다.

Washout wk1.pdf
PDF
Washout wk2.pdf
PDF
Washout wk3.pdf
PDF

좋습니다.

이제 3주 washout 자료까지 완성됐습니다.

오히려 이 자료가 이번 Bioness L300 Go NDIS 승인에서 가장 강력한 근거가 될 수도 있습니다.

제가 확인한 결과, 단순히 "Bioness가 도움이 된다" 수준이 아니라,

Bioness 제거 후 기능 저하(reversal of function)가 매우 명확하게 나타났습니다.

이것은 NDIS에서 매우 좋아하는 evidence입니다.

추가할 수 있는 Washout 결과 분석 (AT Report)
14. Post-Bioness Withdrawal Analysis (3-week washout)

Following completion of the 8-week community trial, the participant returned the Bioness L300 Go and completed a structured 3-week withdrawal period to evaluate changes in function without access to the device.

The participant demonstrated a progressive deterioration in mobility, fatigue, confidence, safety and community participation throughout the washout period.

Emerging themes identified

1. Increased physical fatigue

The participant consistently reported:

severe leg soreness
calf tightness
muscle stiffness
ankle weakness
increased effort required for walking
prolonged recovery periods after activities

Examples included:

"Woke up with extremely sore body and legs especially my right leg."

"Legs very sore and stiff."

2. Increased instability and trips

Near falls and trips re-emerged.

Documented examples included:

Tripped exiting bathroom
Tripped through doorway
Scuffed toes due to inadequate foot clearance
Repeated tripping over own feet

Participant also described:

"Almost completely falling."

and

"Keep tripping over my own feet."

3. Significant reduction in walking quality

Throughout the washout period, the participant repeatedly rated mobility as:

Difficult
Very difficult
Worse
Much worse than when using Bioness

The participant reported:

"Walking was very slow."

"Not very confident."

"Hard to walk."

"Takes me forever to move."

4. Increased cognitive load

Participant described having to consciously think about every movement again.

Without Bioness, mobility became mentally demanding and exhausting.

She reported:

greater concentration required for foot placement
increased awareness of foot position to avoid falls
loss of automatic gait patterning
5. Reduced community participation

Community participation became substantially harder.

Examples included:

difficulty visiting nursing homes
slower movement in community environments
increased rest periods
reduced confidence when leaving home
6. Psychological deterioration

This section is extremely powerful for NDIS.

Participant reported:

"I NEED my Bioness."

"Depressed, unhappy."

"I can't cope with this pain anymore."

Participant reported needing psychology support due to deteriorating mental wellbeing associated with loss of independent mobility.

아주 중요한 결론 문단 (Final Clinical Interpretation)

이 문단은 최종 보고서에 꼭 넣으세요.

Clinical Interpretation

The withdrawal period provides strong evidence that the participant's improvements observed during the 8-week Bioness L300 Go trial were directly attributable to the device rather than natural symptom fluctuation, placebo effect, concurrent physiotherapy intervention, or disease variability.

Upon removal of the device, a clear and progressive decline was observed across multiple domains including walking safety, endurance, fatigue, confidence, balance, community participation and psychological wellbeing.

This pattern strongly supports that ongoing access to the Bioness L300 Go is necessary to maintain the participant's functional capacity and prevent significant deterioration in daily life.

제가 보기에는 지금 가지고 계신 자료 수준이면,

객관적 Outcome Measures + 8주 Community Trial + 3주 Washout + Participant Feedback + Mother Feedback + Neurologist Letter

까지 모두 갖춰져 있어서 상당히 강한 NDIS 제출 자료가 됐습니다.

다음 단계는 이 모든 자료를 통합해서 15~20페이지 분량의 제출용 최종 AT Report(영문 완성본)를 만드는 것입니다.

NDIS_Full_AT_Report_Bioness_L300Go_M.docx
Document
클로드가 만든 문서야. 이렇게 실제 제출할수 있는 형식으로 만들어줘

네. 이게 훨씬 좋네요.

올려주신 Claude 문서를 보니 상당히 잘 만들어져 있습니다.

하지만 NDIS 심사 관점에서 보면 제가 몇 가지 더 보완하고 싶습니다.

현재 문서는 약 90% 완성도입니다.

실제 제출용으로는 조금 더 강화하면 승인 가능성을 높일 수 있습니다.

특히 아래 부분을 수정하면 좋겠습니다.

1. 가장 중요한 수정 사항

현재 문서는

Part A (Community trial 신청 당시 자료)

↓

Part B (Community trial 결과)

구조인데,

실제 NDIS High Cost AT 구매 심사에서는 이제 Community Trial 신청 이유보다 "구매(Purchase Recommendation)" 근거가 더 중요합니다.

따라서 문서 비중을 바꾸는 것을 추천합니다.

현재

Part A : 50%

Part B : 50%

추천

Part A : 30%

Part B : 70%

2. Clinical Analysis를 더 강화

현재 B5는 조금 짧습니다.

아래 내용을 추가하면 좋습니다.

B5.5 Device Dependency Analysis (추가)
B5.5 Device Dependency Analysis

The combination of the 8-week active trial and subsequent 3-week washout period provides particularly strong evidence supporting long-term provision of the Bioness L300 Go.

Unlike a single clinical assessment, this longitudinal evaluation demonstrates that functional improvements are sustainable during active use and regress upon device removal.

Three independent sources of evidence consistently support device dependency:

1. Objective outcome measures

Functional improvements were observed across multiple standardised assessments during active use, with regression during washout.

2. Participant-reported outcomes

The participant consistently reported improved confidence, reduced anxiety, increased endurance and greater willingness to participate in community activities while using the device.

3. Behavioural changes

Activity avoidance, reduced confidence and increased fear of falling re-emerged following withdrawal of the device.

This pattern strongly indicates that the Bioness L300 Go is an essential assistive technology rather than an optional convenience item.

3. NDIS Reasonable & Necessary 부분을 훨씬 강화

현재 표는 조금 짧습니다.

아래처럼 바꾸는 것을 추천합니다.

B7. NDIS Reasonable & Necessary Criteria (Revised)
NDIS Section 34 Criterion Clinical Justification
Related to disability The participant's foot drop is a direct consequence of Multiple Sclerosis-related upper motor neuron dysfunction.
Supports goals and aspirations Directly addresses all four participant goals including mobility, safety, independence and participation.
Effective and beneficial Supported by objective measures, participant feedback, carer feedback, neurologist support and washout analysis.
Value for money Likely to reduce long-term costs associated with falls, injury management, increasing support worker needs and deconditioning.
Takes account of informal supports Enhances independence and reduces reliance on the participant's mother for community access.
Most appropriate support Alternative interventions have either failed or are unable to provide active neuromuscular assistance.
Evidence-based intervention Supported by peer-reviewed evidence and real-world community trial outcomes.
4. 추가해야 하는 섹션 (강력 추천)

현재 문서에는 없는데 넣으면 좋은 섹션입니다.

B7.1 Risks Without Bioness

Without ongoing access to the Bioness L300 Go, the participant is at significant risk of:

Physical Risks
Increased falls
Musculoskeletal injury
Progressive deconditioning

Reduced endurance
Increased fatigue
Functional Risks
Reduced community participation
Reduced independence
Activity avoidance
Psychological Risks
Increased anxiety
Reduced confidence
Social isolation
Reduced self-esteem
Support Risks
Increased reliance on informal supports
Potential escalation in funded supports

These risks were objectively demonstrated during the washout period and are expected to continue without long-term access to the device.

5. Recommendation 페이지를 별도로 분리

현재 Recommendation이 마지막 한 페이지에 모여 있는데, 제출용은 별도 페이지로 만드는 것을 추천합니다.

FINAL RECOMMENDATION FOR NDIS FUNDING

It is the professional opinion of the prescribing Orthotist that the Bioness L300 Go Functional Electrical Stimulation system should be funded as High Cost Assistive Technology under the NDIS.

This recommendation is supported by:

✓ Objective outcome measures

✓ 8-week community trial evidence

✓ 3-week washout findings

✓ Participant feedback

✓ Carer feedback

✓ Neurologist support

✓ Strong alignment with participant goals

✓ Evidence supporting all NDIS Section 34 Reasonable and Necessary criteria

The Bioness L300 Go is not considered an optional device but an essential piece of assistive technology required to safely maintain functional mobility, independence and community participation.

그리고 제가 보기에는 이제 거의 최종 단계입니다.

다음 버전은 단순 보고서가 아니라 실제 Korthotics에서 제출하는 수준으로, 18~20페이지짜리 Professional NDIS High Cost AT Report Version 2.0으로 만드는 것을 추천합니다.

그러면 지금 Claude 문서를 뼈대로 해서 NDIS 플래너나 AT assessor가 읽는 흐름에 맞게 재구성할 수 있습니다.

좋습니다.

이번에는 Claude가 만든 파일을 기본 뼈대로 사용하고, 지금까지 수집된 모든 자료(Initial assessment + In-clinic trial + 8-week community trial + Outcome measures + Participant feedback + Mother feedback + Neurologist support +

3-week washout)를 통합하여 실제 제출용 수준의 NDIS High Cost AT Report를 차례대로 작성해보겠습니다.

그리고 실제 NDIS 심사관이 읽는 순서에 맞춰 구성하겠습니다.

한 번에 20페이지를 작성하면 수정하기 어려우므로, Cover Page → Executive Summary → Participant Profile → NDIS Goals부터 작성하는 것을 추천합니다.

COVER PAGE
NDIS HIGH COST ASSISTIVE TECHNOLOGY ASSESSMENT REPORT
Funding Request for the Bioness L300 Go Functional Electrical Stimulation (FES) System
Supporting Ongoing Provision Under the National Disability Insurance Scheme (NDIS)
Participant Details
Item Details
Participant Name __
NDIS Number __
Date of Birth __
Primary Diagnosis Multiple Sclerosis
Date of Report __
Prescribing Clinician
Item Details
Assessor __
Profession Orthotist
Qualifications __
Organisation Korthotics
Contact Details __
Supporting Evidence Included Within This Report

✓ Comprehensive Orthotic Assessment

✓ In-Clinic Bioness L300 Go Trial

✓ 8-Week Community Trial

✓ 3-Week Washout Period

✓ Objective Outcome Measures

✓ Participant Logbooks

✓ Participant Feedback

✓ Mother/Carer Feedback

✓ Treating Neurologist Support

✓ NDIS Reasonable and Necessary Analysis

TABLE OF CONTENTS

1. EXECUTIVE SUMMARY
   Purpose of Report

This report has been prepared to support funding for the Bioness L300 Go Functional Electrical Stimulation (FES) system as High Cost Assistive Technology under the National Disability Insurance Scheme (NDIS).

This recommendation is based on comprehensive evidence obtained through a multi-stage assessment process, including:

Comprehensive orthotic assessment
Positive in-clinic trial
Eight-week community trial
Three-week washout period
Standardised outcome measures
Participant logbooks
Participant feedback
Carer observations
Treating neurologist support

The purpose of this report is to demonstrate that the recommended assistive technology satisfies the NDIS Act 2013 Section 34 Reasonable and Necessary criteria.

Summary of Findings

M is a 35-year-old woman with a confirmed diagnosis of Multiple Sclerosis (MS) who experiences significant right-sided neurological impairment resulting in:

Severe foot drop
Gait instability
Increased falls risk
Reduced endurance
Significant fatigue
Reduced confidence with mobility
Reduced participation in community activities

These impairments substantially limit her independence and ability to safely participate in daily life.

An initial in-clinic trial demonstrated immediate improvements in gait quality, foot clearance, walking speed and fatigue management.

The participant subsequently completed an eight-week community trial within her everyday environments.

During this period, meaningful improvements were consistently demonstrated across multiple functional domains including:

Walking efficiency
Endurance
Confidence
Community participation
Independence
Safety

Objective outcome measures also demonstrated measurable improvement.

A subsequent three-week washout period was then undertaken following removal of the device.

During this period, deterioration occurred across multiple domains including:

Increased fatigue
Increased tripping
Reduced confidence
Reduced participation
Reduced walking efficiency

The deterioration observed following device withdrawal strongly supports that the improvements obtained during the community trial were directly attributable to the Bioness L300 Go rather than disease fluctuation, placebo effect, or concurrent therapies.

Based on the totality of evidence, long-term provision of the Bioness L300 Go is strongly recommended as reasonable and necessary Assistive Technology under the NDIS.

2. PARTICIPANT PROFILE
   Item Details
   Age 35 years
   Primary Diagnosis Multiple Sclerosis
   Primary Impairment Right-sided foot drop
   Mobility Aid Four-wheeled walker
   Living Arrangement __
   Informal Supports Family support
   Disability Impact

The participant experiences permanent neurological impairments affecting her mobility, endurance, safety and independence.

Her impairments fluctuate daily due to the nature of Multiple Sclerosis but remain persistent and progressive.

Primary functional limitations include:

Inability to actively dorsiflex the right ankle
Reduced foot clearance during gait
Right ankle inversion during swing phase
Right knee hyperextension during stance phase
Increased fatigue during walking activities
Reduced walking endurance
Reduced confidence with community access
Increased falls risk

These impairments substantially impact her ability to safely participate in everyday life.

3. PARTICIPANT GOALS AND ASPIRATIONS

The proposed Bioness L300 Go directly aligns with the participant's identified NDIS goals.

Participant Goal How the Bioness L300 Go Supports This Goal
Increase mobility, strength and stability to reduce falls risk and increase independence Improves gait efficiency, foot clearance, endurance and stability
Improve mobility to support overall health Enables increased physical activity and participation
Be safe within the home Reduces tripping risk and improves walking safety
Increase independence and coping skills Improves confidence, participation and self-management

The participant demonstrated measurable progress toward each of these goals throughout the community trial.

다음 단계는 실제 보고서의 핵심인 Section 4~7 (Medical History, Functional Capacity, Alternative Solutions, Clinical Assessment Findings)를 작성하는 것입니다.

이 부분부터 NDIS 심사관들이 본격적으로 보는 임상 근거가 시작됩니다.

다음단계를 NDIS 심사관이 봤을때 필요하다는 것을 확실이 알수 있도록 자료를 기반으로 만들어줘

좋습니다.

지금부터는 Orthotist가 작성하는 Clinical Report가 아니라 NDIS Delegate(심사관)를 설득하는 Funding Justification document라고 생각하고 작성해야 합니다.

심사관은 의료인이 아닐 수도 있기 때문에 단순히 평가 결과를 나열하면 안 됩니다.

항상 아래의 논리 구조로 작성해야 합니다.

Permanent impairment → Functional limitation → Failed alternatives → Proven benefit → Deterioration without device → NDIS funding is required

특히 현재 가지고 있는 자료의 가장 큰 강점은 3주 washout입니다.

이걸 적극적으로 활용해야 합니다.

4. Medical History and Disability Impact
   4.1 Medical Background

M is a 35-year-old participant with a confirmed diagnosis of Multiple Sclerosis (MS), a progressive neurological condition causing permanent and fluctuating impairments that significantly impact her mobility, safety and ability to participate independently in everyday life.

M experiences substantial right-sided lower limb dysfunction which has progressively reduced her ability to safely mobilise both within her home and in the community.

Although symptoms fluctuate from day to day, her neurological impairments are permanent and are expected to continue to impact her functional capacity long-term.

These impairments are directly attributable to her disability and cannot be resolved through rehabilitation alone.

The participant remains highly motivated to maintain her independence and actively participates in physiotherapy and regular exercise; however, despite these efforts, significant mobility limitations persist.

4.2 Primary Disability Related Impairments

Clinical assessment identified the following impairments.

Muscle Strength
Muscle Group Right Side

Ankle Dorsiflexion 0/5
Ankle Plantarflexion 3/5
Ankle Eversion 0/5
Ankle Inversion 1/5
Knee Flexion/Extension 4/5
Hip Abduction/Extension 5/5

Additional findings included:

Ankle dorsiflexion ROM with knee extended: approximately -3°
Moderate spasticity of gastrocnemius and soleus
Mild spasticity of ankle invertors
Toe curling with prolonged standing and walking

Gait assessment identified:

Significant right foot drop during swing phase
Right ankle inversion during swing phase
Right knee hyperextension during stance phase
Reduced gait efficiency
Increased energy expenditure

These impairments significantly increase the participant's risk of falls and substantially reduce her ability to mobilise safely and efficiently.

5. Current Functional Capacity and Disability Impact

Despite significant effort to maintain mobility, M experiences considerable functional limitations throughout her daily life.

Home Mobility

The participant mobilises independently indoors using a four-wheeled walker.

However, mobility remains slow, effortful and cognitively demanding.

She must consciously monitor every step in order to avoid tripping due to inadequate foot clearance.

Activities that most individuals complete automatically require substantial physical and mental effort.

The participant experiences increased fatigue even during routine household mobility.

Community Mobility

Community access is significantly restricted.

The participant experiences difficulty with:

Walking on uneven surfaces
Negotiating ramps
Walking long distances
Attending appointments
Shopping activities
Participating in social activities
Completing multiple tasks within one outing

Fatigue often accumulates throughout the day and further limits participation.

As a result, the participant frequently self-limits activities to preserve energy and reduce the risk of falling.

This significantly impacts her independence and quality of life.

Fatigue

Fatigue is one of the participant's most disabling symptoms.

The participant demonstrated severe fatigue affecting multiple domains of daily function.

Fatigue directly impacts:

Mobility
Endurance
Participation
Confidence
Social engagement
Emotional wellbeing

As fatigue increases throughout the day, gait quality deteriorates, further increasing the risk of trips and falls.

Participation Restrictions

Without appropriate assistive technology support, the participant experiences reduced participation in meaningful life activities.

This includes:

Reduced ability to independently attend appointments
Reduced ability to participate in family activities
Reduced confidence leaving home
Reduced ability to access community environments
Reduced ability to sustain prolonged activities

These participation restrictions place the participant at risk of progressive social isolation.

6. Previous Interventions and Alternative Solutions Considered

Several interventions have been trialled to address the participant's foot drop and mobility limitations.

Physiotherapy

The participant actively engages in physiotherapy and remains committed to maintaining function.

While physiotherapy is beneficial for strength maintenance and overall conditioning, it has been insufficient to overcome the significant neurological deficit affecting ankle dorsiflexion.

Physiotherapy alone cannot compensate for absent dorsiflexion during walking.

Off-the-Shelf AFO (OAPL Supra-Lite)

The participant trialled an off-the-shelf OAPL Supra-Lite AFO under physiotherapy guidance.

This intervention was unsuccessful due to:

Poor fit
Bulkiness
Reduced comfort
Presence of spasticity
Reduced acceptance by the participant

The participant was unable to consistently integrate the device into everyday life.

Custom AFO

A custom AFO has been explored as a potential option.

Whilst a custom AFO may improve foot clearance, it remains a passive orthotic intervention.

A custom AFO cannot provide:

Active muscle recruitment
Dynamic neuromuscular assistance
Therapeutic benefits associated with repetitive stimulation
Adaptability to fatigue fluctuations associated with Multiple Sclerosis

As such, a custom AFO is unlikely to provide the same level of functional benefit demonstrated during the Bioness trial.

Comparative Analysis of Alternative Solutions
Intervention Outcome Reason Not Recommended
Physiotherapy only Insufficient Cannot replace absent dorsiflexion
OAPL Supra-Lite AFO Unsuccessful Poor fit and poor acceptance
Custom AFO Partial solution Passive support only
No intervention Unsafe Increased falls risk and deterioration
Bioness L300 Go Successful Demonstrated significant real-world benefits

The Bioness L300 Go was the only intervention that consistently improved function across multiple domains and was successfully integrated into the participant's daily life.

7. Why This Device Is Necessary (VERY IMPORTANT SECTION)

The Bioness L300 Go is not being recommended because it is a technologically advanced device.

It is being recommended because it was the only intervention that demonstrated sustained, measurable and reproducible improvements in real-world function.

Unlike a single clinical assessment, the participant completed:

In-clinic assessment
8-week community trial
3-week washout period

This longitudinal evaluation demonstrated that:

During active use:

✓ Walking improved

✓ Endurance improved

✓ Confidence improved

✓ Community participation improved

✓ Safety improved

Following device removal:

✗ Tripping increased

✗ Fatigue increased

☓ Confidence declined

☓ Participation reduced

☓ Emotional wellbeing deteriorated

This pattern strongly indicates that the participant has developed a functional dependency on the device to safely maintain her mobility and independence.

The Bioness L300 Go should therefore be considered essential assistive technology rather than an optional convenience item.

제가 보기에는 다음에 작성할 Section 8~11 (In-clinic Trial, Community Trial Analysis, Outcome Measures, Washout Analysis)가 이 보고서의 당락을 결정하는 핵심 파트입니다.

그 부분에서 NDIS 심사관이 "이건 사줘야겠다"라고 판단할 정도의 논리 구조를 만들어야 합니다.

핵심파트로 자료를 기반으로 만들어줘

좋습니다.

이제부터는 NDIS 심사관의 승인 여부를 결정하는 핵심 파트 (Section 8~11) 를 작성하겠습니다.

여기서 가장 중요한 것은 "Bioness가 도움이 됐다"를 반복하는 것이 아니라, 기존 중재로 해결되지 않았던 장애로 인한 기능 제한을 Bioness가 해결했고, 제거하자 다시 악화됐다는 인과관계(causality)를 증명하는 것입니다.

그리고 제가 지금까지 주신 자료를 보면 가장 강력한 4가지 증거가 있습니다.

① In-clinic trial

② 8-week community trial

③ Objective outcome measures

④ 3-week washout

이 네 가지를 하나의 스토리로 연결해야 합니다.

SECTION 8. In-Clinic Trial Findings and Clinical Suitability Assessment

M attended Korthotics on 24 September 2025 accompanied by her mother and treating physiotherapist to determine suitability for Functional Electrical Stimulation (FES) using the Bioness L300 Go system.

The purpose of the assessment was to determine whether active neuromuscular assistance could address the participant's significant right-sided foot drop, which had not been adequately managed through previous interventions.

The participant underwent a supervised clinical trial in which stimulation parameters were optimised to facilitate ankle dorsiflexion during gait.

Immediate functional improvements were observed.

Improvements observed during the assessment included:
Gait quality
Improved foot clearance during swing phase
Reduction in foot dragging
Reduction in ankle inversion during gait
Stability
Improved stance stability
Reduced compensatory movement patterns

Walking efficiency
Increased walking speed
Reduced energy expenditure during walking
Fatigue management
Reduced perceived effort during mobility activities

The participant tolerated the device well and demonstrated an immediate positive response.

These findings indicated that the participant was clinically suitable for a longer-term community trial.

However, given the fluctuating nature of Multiple Sclerosis, a short clinical assessment alone was considered insufficient to justify long-term prescription.

A real-world community trial was therefore required to determine whether these benefits could be sustained in everyday environments.

SECTION 9. Eight-Week Community Trial Analysis
Purpose of the Community Trial

The participant completed an eight-week community trial to determine whether the benefits observed during the initial assessment could be translated into meaningful and sustainable improvements in daily life.

Unlike a clinic environment, the community trial allowed assessment of the participant's performance across multiple environments, varying fatigue levels and real-life demands.

Throughout the trial, the participant completed structured logbooks documenting her experience.

She incorporated the device into everyday activities including:

Home mobility
Shopping
Medical appointments
Hairdresser appointments
Family outings
Outdoor walking
Physiotherapy sessions
Household activities
Key Functional Improvements Observed
Improved Mobility

Throughout the trial period, the participant consistently reported improved foot clearance and reduced foot dragging.

She described being able to lift her foot more easily and move with greater efficiency.

This reduced the amount of concentration required to walk safely.

Improved Confidence

Repeated logbook entries demonstrated increased confidence when mobilising both indoors and outdoors.

Confidence improved when negotiating:

Uneven surfaces
Ramps
Driveways
Outdoor environments

This confidence translated into increased willingness to leave the home and participate in meaningful activities.

### Increased Community Participation

The participant progressively increased her participation in community activities throughout the trial.

Activities included:

Attending appointments
Shopping
Hairdresser visits
Family activities
Outdoor recreational walking

These activities represent meaningful participation outcomes rather than isolated mobility tasks.

### Improved Endurance

The participant demonstrated increased tolerance for sustained activity throughout the trial period.

This occurred despite the underlying progressive nature of Multiple Sclerosis.

### Safety Outcomes

Importantly, no falls were reported throughout the community trial period whilst actively using the device.

## SECTION 10. Objective Outcome Measure Analysis

Objective outcome measures were completed at four separate time points.

Pre-trial baseline
Mid-trial (4 weeks)
End of trial (8 weeks)
Post-trial washout

These measures were selected to assess mobility, endurance, balance and functional independence.

### Outcome Measure Results

| Outcome Measure | Pre | Mid | End | Washout |
|---|---|---|---|---|
| Timed Up and Go | 42.57 sec | 43.52 sec | 33.11 sec | 35.42 sec |
| 30 Second Sit to Stand | 9 | 9 | 10 | 9 |
| 10 Metre Walk Test | 35.19 sec | 28.5 sec | 29.31 sec | 31 sec |
| 6 Minute Walk Test | 57 m | 60 m | 90 m | 75 m |
| Dynamic Gait Index | 11/24 | 14/24 | 14/24 | 12/24 |

### Clinical Interpretation

#### Timed Up and Go (TUG)

TUG improved from 42.57 seconds to 33.11 seconds by the end of the trial.

This represents a 22% improvement in functional mobility, demonstrating improved walking efficiency and safer transitional movements.

Following withdrawal, performance deteriorated to 35.42 seconds.

#### 6 Minute Walk Test (6MWT)

The participant improved from 57 metres to 90 metres.

This represents a 58% increase in walking endurance.

Following removal of the device, distance reduced to 75 metres.

This demonstrates reduced functional capacity without ongoing access to the device.

Dynamic Gait Index (DGI)

Dynamic balance improved from 11/24 to 14/24.

Following washout, this reduced to 12/24.

This pattern indicates improved balance and reduced falls risk during active device use.

Overall Interpretation

The participant demonstrated meaningful improvements across multiple objective measures whilst using the device.

Regression occurred following withdrawal.

This pattern strongly supports a causal relationship between device use and improved functional outcomes.

SECTION 11. Washout Analysis and Device Dependency
Purpose of the Washout Period

Following completion of the community trial, the participant returned the device and completed a three-week washout period.

The purpose of this process was to determine whether the improvements observed during the trial were sustained independently or whether they were dependent upon ongoing access to the device.

This analysis is particularly important given the fluctuating nature of Multiple Sclerosis.

Findings

A clear and progressive deterioration was observed.

Physical deterioration

The participant reported:

Increased leg soreness
Increased calf tightness
Increased stiffness
Increased fatigue
Mobility deterioration

The participant reported:

Slower walking
Increased effort required for mobility
Increased foot dragging
Safety deterioration

The participant experienced:

Increased tripping
Increased toe scuffing
Near-fall episodes
Participation deterioration

The participant reduced activity levels and required more frequent rest periods.

Psychological deterioration

The participant reported:

"I NEED my Bioness."

"I can't cope with this pain anymore."

She also described feeling depressed, unhappy and less able to manage her condition.

Clinical Interpretation

The deterioration observed during the washout period provides strong evidence that the functional improvements observed throughout the community trial were directly attributable to the Bioness L300 Go.

The participant demonstrated improvement during active use and deterioration during withdrawal.

This pattern was consistently observed across:

Objective outcome measures
Participant logbooks
Participant feedback
Carer observations

This pattern cannot be explained by placebo effect, natural disease fluctuation or concurrent therapies alone.

It is therefore the professional opinion of the prescribing Orthotist that ongoing access to the Bioness L300 Go is required to maintain the participant's current level of functional capacity, safety and independence.

다음 단계(Section 12~15)가 실제 승인 여부를 결정하는 마지막 부분입니다.

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